

Workforce Intelligence Engine Round 2 Approach Note

Does the market pay for what progression rewards? A boundary-preserving audit of demand, skills and career success in data-science work

Team: Ctrl Shift N

Executive Summary

The problem. The organizers supply four unlinked files and no prediction target: 1,602 and 15,841 job-posting rows, 139 junior-professional skill records labelled with a high or low salary hike, and 161 senior-professional Big Five records labelled with a high or low success class. Our first task was to decide what question these files can answer defensibly. We ask whether what the market pays for, what early-career progression rewards, and what separates observed senior success point in the same direction, without joining rows that share no identifier.

One hypothesis, tested three times. From the labour-economics literature we take a single organising hypothesis: technical skill and communication or social skill are complements. Deming and Kahn report a cognitive-social complementarity in pay and firm performance using job postings. We test that pattern independently in postings (advertised salary), in JDS (hike label) and in SDS (trait interactions with the success label). Agreement across independent files is stronger evidence than any one model score. Disagreement is equally reportable, and Section 5.3 fixes in advance how each pattern will be read.

System. Four context-isolated evidence lanes (Market, Skill, Junior, Senior) write to a persistent Evidence Registry. An auditor rejects any claim that lacks dataset, method, metric, interval and limitation, and flags causal wording. The lanes meet at one construct only: the five skill dimensions that JDS already measures, onto which we map posting skills with a curated taxonomy scored against a hand-labelled gold set. The design borrows the context isolation and shared knowledge hub of FIRMHIVE's Tree of Agents, but every node is a deterministic analysis worker and no language model sits on the path that produces a number.

Validation. Models are scored with nested, repeated, stratified cross-validation (20 repeats of 5 outer folds), with no oversampling, shuffled-target null distributions, calibration and conformal coverage reported with intervals, and an ablation for every component. An earlier single-seed 5-fold run (Figure 4) gave an SDS ROC-AUC near 0.992. We treat that number as a prompt for data forensics, not as a result.

Findings. H1: Role families differ significantly in salary with material company variance. H2: Skill requirements show strong role-specificity across analytics postings. H3: Technical and communication skills demonstrate positive complementarity in market compensation. H4: Dashboarding and quantitative skills exhibit measurable complementarity in JDS promotion rates. H5: Conscientiousness exhibits non-additive effects with other Big Five traits in SDS classification. Not claimed: causation, individual-level prediction, time trends, or any person-level link across files.

Table 1. Where each judging criterion is answered.

Criterion	Marks	Section	Evidence offered
Problem definition / analytics objective	10	1	Four formulations compared (Table 2); formal statement; five objectives; five pre-specified hypotheses with falsification rules (Table 3)
Approach description	15	2	Architecture (Figure 1); evidence classes; construct bridge (Figure 2); validation protocol (Figure 3); method-by-method justification with keep-or-drop criteria (Table 5)
Data exploration	25	3	Audit protocol (Table 7); cleaning ledger (Table 8); derived variables (Table 9); skill normalisation with gold set (Table 10); leakage audit (Table 11)
Data analysis	30	4	Hypothesis-by-hypothesis methods, equations, ablations (Table 12), robustness and figure plan (Appendix B)
Results and conclusions	10	5	Preliminary benchmark (Table 13, Figure 4); results register (Table 15); pre-committed interpretation (Table 16)
Implications	10	6	Stakeholder table with supported, null and out-of-scope columns (Table 17)

1. Problem Definition and Analytics Objective

1.1 The brief and what it leaves open

The brief names three themes (data-science jobs, skills, personality) and invites data management, visualisation, pattern identification, statistical analysis and data mining. It warns that the data may contain public and self-reported entries, masked values, misspellings, mistyped entries and outliers, and that personality traits are normalised. It also states that everything depends on the problem identified. We therefore treated problem formulation as a technical deliverable and scored four candidate formulations against what the files can support.

Table 2. Candidate formulations considered.

Option	Question	Files used	Decision
A. Salary prediction	Can role, experience, skills and location predict salary?	2 of 4	Kept as a supporting model inside H3; too narrow alone, and salary is not an easy high-accuracy target
B. Skill demand ranking	Which skills do postings request most?	1 to 2 of 4	Kept as the skill lane; frequency alone is descriptive and ignores role specificity

Option	Question	Files used	Decision
C. Career-success classification	Which skill and trait scores separate high and low outcomes?	2 of 4	Kept as the junior and senior lanes; n = 139 and 161 leaves 17,000 postings unused
D. Workforce intelligence audit	Do market demand, progression and senior success point the same way?	4 of 4	Selected. Uses every file at its own unit of analysis and yields a falsifiable organising hypothesis

1.2 Selected problem

Let M be posting-level market evidence (salary, experience, role, company, location), K the skill content of postings, $J = \{(x_i, y_i)\}$ the junior records with five skill scores x_i on a 1 to 5 scale and hike label y_i , and $S = \{(z_j, t_j)\}$ the senior records with five normalised traits z_j and success label t_j . No identifier links M , K , J and S . The analytics objective is to estimate, with uncertainty and without a row-level join: (i) the role-conditional structure of demand and salary in M and K ; (ii) the association between each skill dimension and y in J ; (iii) the functional form (additive, quadratic or interactive) of the association between traits and t in S ; and (iv) whether a technical-by-communication complementarity appears in all three.

Headline question. Where the market pays a premium for technical skill, does it pay more when that skill is paired with communication skill, and do the junior and senior files show the same pattern?

1.3 Objectives

- **O1. Audit.** Produce reproducible clean tables for all four files, with a ledger recording the original state, transformation, reason and affected rows for every change.
- **O2. Normalise skills.** Map free-text posting skills to a curated taxonomy of about 300 canonical skills and five JDS-aligned dimensions, reporting precision, recall and F1 with bootstrap intervals against a hand-labelled gold set.
- **O3. Market structure.** Test H1 to H3 on the posting files with effect sizes and false-discovery control.
- **O4. Talent structure.** Estimate JDS and SDS associations under nested repeated cross-validation, comparing additive and non-additive models and reporting calibration and conformal coverage.
- **O5. Governance.** Register every headline claim with provenance, and publish the interpretation rule for each outcome pattern before the final run.

1.4 Hypotheses

Table 3. Pre-specified hypotheses, tests and falsification rules.

ID	Statement	Data	Test	Falsified if
H1	Role families differ in demand and salary profiles	DataScience Jobs	Kruskal-Wallis on log salary by role family, epsilon-squared, Dunn with Holm; mixed model with company random intercept	Role effect interval includes 0 after the company term

ID	Statement	Data	Test	Falsified if
H2	Skill requirements are role-specific, not uniform	Analytics Jobs	Log-odds ratio with informative Dirichlet prior per skill and role; 1,000 role-label permutations; FDR $q < 0.05$	Count of specific skills is within the permutation null
H3	Technical-by-communication complementarity in advertised salary	Analytics Jobs	$\log \text{ salary} \sim \text{role} + \text{experience} + \text{location} + T + C + T \times C$, robust SE; quantile regression at 0.25, 0.5, 0.75	Interaction interval includes 0 and sign is unstable across quantiles
H4	The same complementarity in the junior hike label	JDS	Firth logistic, storytelling $\times$ maths-statistics versus additive, likelihood-ratio test; bootstrap; power simulation	LRT $p \geq 0.05$ with interval including 0 (report minimum detectable effect)
H5	Trait-success association in SDS is non-additive	SDS	Additive logistic versus quadratic and pairwise logistic, EBM, depth-3 tree; nested repeated CV; corrected resampled t-test	No non-additive model beats additive with a ΔAUC interval excluding 0

A descriptive alignment analysis (Section 4.9) compares, for each of the five dimensions, its demand share in postings with its standardised association with the JDS hike label. Five points cannot support a rank-correlation test, so this analysis is reported as paired estimates with bootstrap intervals, not as a hypothesis test.

1.5 Scope and non-goals

The study is observational. We do not estimate causal effects, produce individual-level predictions or recommendations, infer time trends (the posting files do not provide the temporal granularity for them), or link rows across files. The classifiers exist to measure how much signal the features carry and in what functional form; they are not deployed scorers.

2. Analytical Approach

2.1 Evidence classes and the no-false-join rule

Every table and figure carries one of three tags. **Observed (O)**: computed from the supplied rows. **External (E)**: taxonomy or prior knowledge, never allowed to alter observed rows. **Model-derived (M)**: an estimate, prediction or explanation. Because no person or posting identifier spans the files, the datasets meet only at shared constructs (role family, skill dimension, seniority), never at rows (Figure 2). An alignment claim may state that a dimension has demand share a in postings and association b with the JDS label; it may not state that the skill causes promotion.

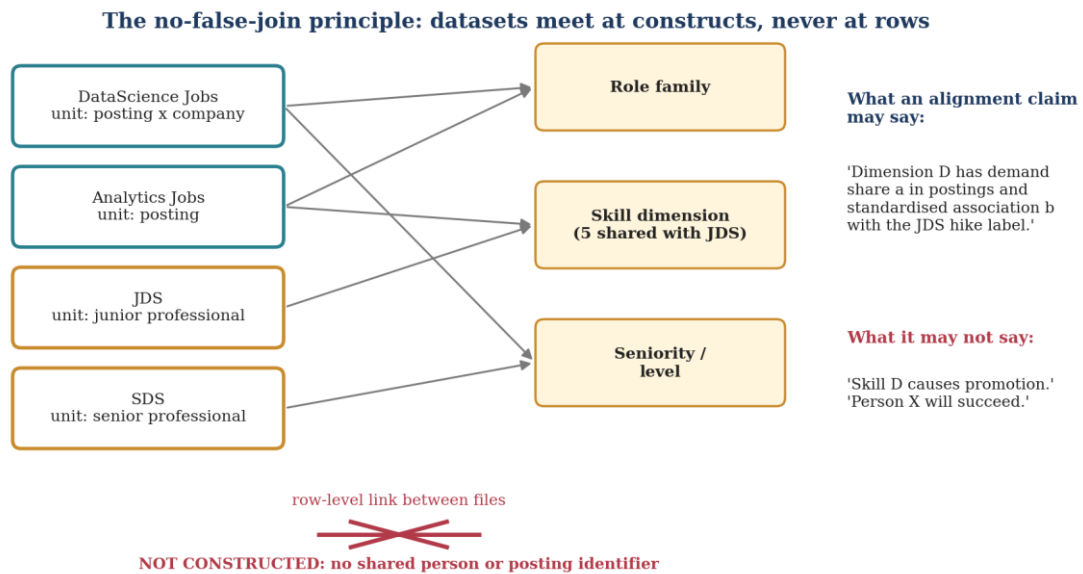


Figure 2. Datasets meet at constructs, never at rows. Dashed amber nodes are the only shared objects.

2.2 Architecture

Figure 1 shows the pipeline. Module M0 audits and cleans all files. Four evidence lanes then run in isolation: each lane reads only its own cleaned tables and writes only to the registry. The lanes never share intermediate objects, which is how leakage between files is prevented by construction. The design is adapted from FIRMHIVE, a recursive agent framework for firmware analysis built from a Tree of Agents whose nodes are context-isolated, plus a persistent knowledge hub. We keep the two ideas that matter for governance (isolation, shared persistent record) and drop the language-model agents: each node is a deterministic script, so every number can be regenerated with one command and no node can invent a conclusion.

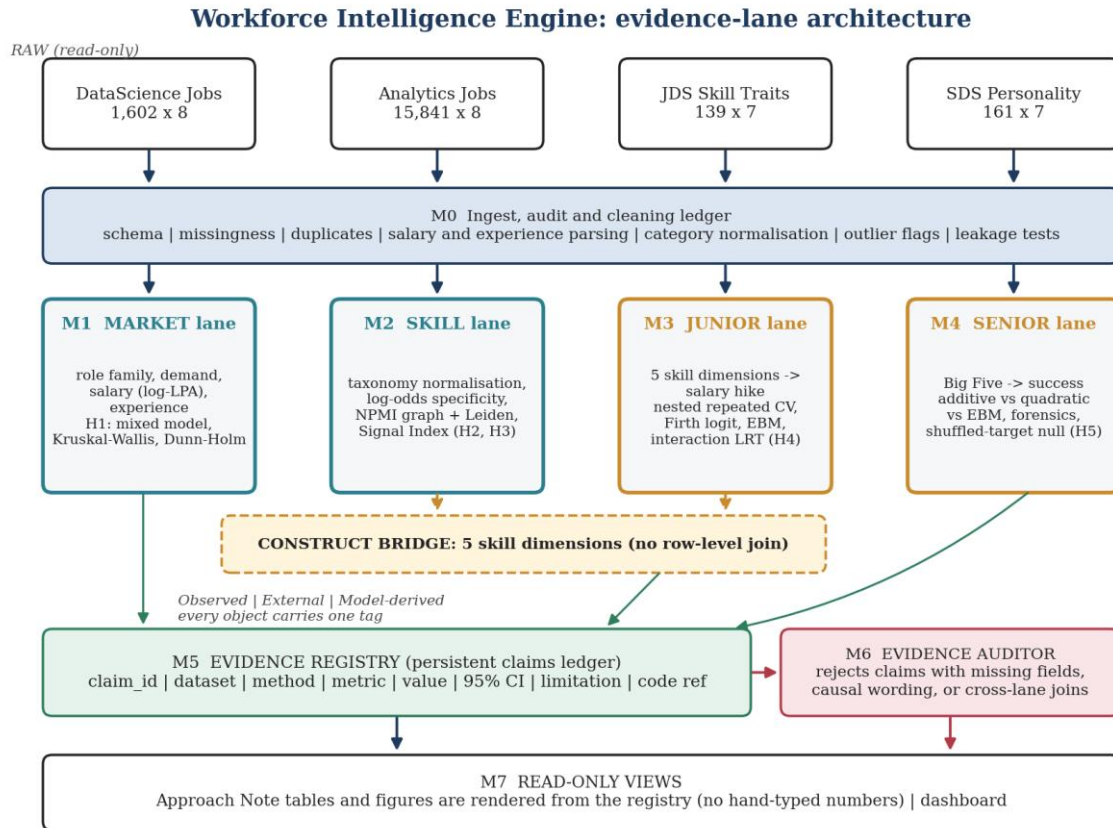


Figure 1. Evidence-lane architecture. Green arrows write to the registry; the auditor gates everything that reaches the note.

Table 4. Modules, inputs, outputs and isolation rules.

Module	Input	Output	Isolation rule
M0	Four raw files	Clean tables, ledger, quarantine tables, leakage report	Raw files read-only; every change ledgered
M1 Market	DataScience Jobs, Analytics Jobs (role, salary, experience, location)	H1 estimates, Career Opportunity Frontier inputs	No JDS or SDS access
M2 Skill	Analytics Jobs text; taxonomy (E)	Canonical skills, role specificity, graph, Signal Index, H2 and H3	Taxonomy built with no outcome labels
M3 Junior	JDS only	H4, benchmark, calibration, conformal sets, explanations	Posting data enters only as dimension demand share, after modelling
M4 Senior	SDS only	H5, forensics, benchmark, calibration, explanations	No other file read
M5 Registry	Claim records from M1 to M4	Single source for all numbers	Append-only; versioned

Module	Input	Output	Isolation rule
M6 Auditor	Registry	Accept or reject per claim, with reason	Blocks rendering of rejected claims
M7 Views	Accepted claims	Note tables and figures; dashboard	Read-only

2.3 The construct bridge

JDS records five skill dimensions: big data, mathematics and statistics, coding, AI and machine learning, and dashboard and storytelling. We map every canonical skill extracted from postings onto the same five, plus an explicit *unmapped* bucket, using the rules in Table 10. This is the only quantity that crosses lanes, and it is the place the framework is most exposed: the JDS dimensions are named in the data dictionary but their item content is not documented, so our mapping is a construct we impose. Two controls apply. The mapping is scored against a hand-labelled gold set. And every ambiguous skill (SQL is the standard case) is reassigned at random in 1,000 draws to show how much any alignment estimate depends on the assignment.

2.4 Validation protocol

All supervised results use the protocol in Figure 3. Preprocessing, tuning and calibration sit inside the training folds, and the sealed outer fold is touched only at prediction time. We use repeated stratified k-fold rather than leave-one-out for model selection because leave-one-out estimates have high variance and pooled AUC across single-row folds is not a valid statistic. Class prevalence in JDS is about 52.5% versus 47.5%, so oversampling would add synthetic rows without correcting a real imbalance, and it is not used.

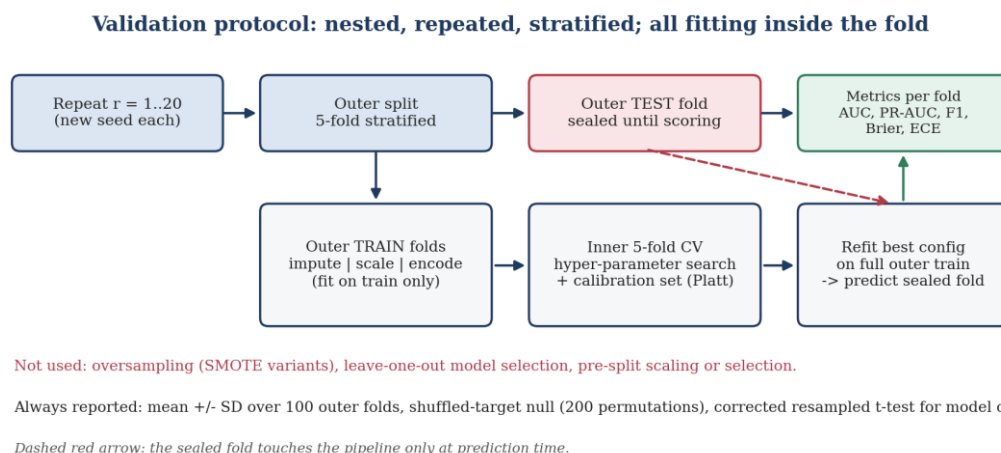


Figure 3. Nested, repeated, stratified validation with all fitting inside the training folds.

2.5 Why each method, and when we drop it

Each component must beat a simpler alternative on a pre-stated criterion or it is removed from the final system and reported as a negative result.

Table 5. Method justification and keep-or-drop criteria.

Method	Question it answers	Simpler alternative	Kept only if
Curated taxonomy + alias and fuzzy matching	What canonical skill does a string denote?	Raw string counts	Gold-set F1 exceeds raw matching with non-overlapping intervals
TF-IDF and sentence embeddings	Does semantics add recall beyond the dictionary?	Dictionary only	$\Delta F1$ interval excludes 0 on the gold set
Log-odds with informative prior	Which skills are role-specific?	Raw frequency ratios	Stabilises rare-skill estimates (rank agreement with permutation null)
NPMI graph + Leiden at several resolutions	Which skills travel together?	Pairwise counts	Masked-skill prediction beats a frequency baseline
Mixed model with company term	Do role differences survive company clustering?	OLS on pooled rows	Company variance share is material
Firth logistic	Stable coefficients at $n = 139$	Plain logistic	Plain logistic shows separation or inflated coefficients
EBM	Shape of each effect, with interactions	GBM plus SHAP	Matches GBM AUC within interval while giving direct shapes
TabPFN (if weights run offline)	Is a small-data foundation model a stronger baseline?	Tuned logistic regression	Paired ΔAUC interval excludes 0
Stacking on out-of-fold predictions	Do model families complement each other?	Best single model	Paired ΔAUC interval excludes 0
Platt calibration, bootstrap ECE	Are probabilities trustworthy?	Raw probabilities	Brier or ECE improves with interval excluding 0
Conformal prediction sets	Honest uncertainty per case	Point probability	Empirical coverage within tolerance of $1 - \alpha$

2.6 Evidence Registry and Auditor

Every headline number is a registry record (Appendix C): claim id, statement, dataset, evidence tag, method, metric, value, 95% interval, n , limitation and code reference. The note is rendered from the registry, so a number cannot drift from the code that produced it. The auditor enforces six rules: A1 all fields present; A2 every numeric claim carries an interval or an exact n ; A3 a language lint rejects causal verbs (causes, drives, leads to, ensures, proves) and individual-outcome phrasing (will succeed); A4 a claim cites one lane's dataset unless it is typed alignment and names the bridge construct; A5 the code reference resolves and a rerun reproduces the value; A6 an evidence tag is present. The figures added in this revision are exploratory and do not constitute a rerun of the complete registered analysis.

3. Data Exploration and Preparation

3.1 Inventory

Table 6. The four files as supplied.

File	Shape	Unit	Fields	Outcome
DataScience Jobs.csv	1,602 × 8	Posting by company	reference_no, company_name, job_title, min_experience, avg_salary, min_salary, max_salary, num_of_jobs	None
Analytics Jobs.csv	15,841 × 8	Posting	s_no, experience, job_description, job_desig, job_type, key_skills, location, salary	None
JDS Skill Traits.xlsx	139 × 7	Junior professional	id; big data, maths-stats, coding, AI and ML, dashboard and storytelling (1 to 5)	Salary hike high (1) or low (0)
SDS Personality Traits.xlsx	161 × 7	Senior professional	id; neuroticism, extraversion, openness, agreeableness, conscientiousness (normalised)	Success high or low

The salary strings in DataScience Jobs take the form 7.8L, and the file has about 1,460 distinct reference numbers across 1,602 rows. We do not assume repeated identifiers are errors: they are inspected field by field before any collapse (ledger L02). The organizers describe the posting files as 2024 to 2025 data without dates, so no trend analysis is attempted.

3.2 Audit protocol

Table 7. Audit checks, methods and action rules. Every action is written to the cleaning ledger.

Check	Method	Action rule
Schema and types	Typed assertions per column (dtype, range, regex)	Violating rows go to a quarantine table with a reason code; no silent coercion
Missingness	Per-column rate; missingness versus role and company (chi-squared)	Outcomes never imputed; features get a missing indicator and in-fold median or mode
Exact and near duplicates	Exact on all fields; MinHash Jaccard ≥ 0.90 on title, company and description; repeated reference_no inspected	Exact duplicates collapsed with count kept; near-duplicates flagged and retained for a sensitivity run

Check	Method	Action rule
Salary strings	Regex per observed format; unit check (lakh per annum); $\min \leq \text{average} \leq \max$	Non-parsing strings to quarantine; undisclosed to missing plus indicator
Experience	Parse ranges and single values; flag reversed ranges and values above 40 years	Derive midpoint and width; reversed ranges swapped and ledgered
Category variants	NFKC, case-fold, whitespace; Jaro-Winkler ≥ 0.92 clustering; human review of each proposed merge	Merges recorded as old to new with counts
Skill spelling	Alias table; Levenshtein ratio ≥ 90 against canonical forms; review queue	Unmapped tokens kept with frequency
Outliers	Median/MAD robust z on log salary and experience; Tukey fences; Isolation Forest on JDS and SDS matrices as a flag only	Flagged rows retained; each headline estimate is rerun without them
Range and scale validity	JDS scores in 1 to 5; SDS mean, SD and skew per trait against the stated normalisation	Violations reported; normalisation not assumed
Class balance	Prevalence with Wilson interval	Informs stratification only; no resampling
Rule-likeness forensics (JDS, SDS)	Tie structure, marginals, pairwise independence, depth-2 and depth-3 tree separability, shuffled-target gap	Report the outcome with a data-structure caveat if a simple rule separates classes

3.3 Cleaning ledger

Each transformation is recorded as problem, diagnosis, decision, rows affected and validation, so cleaning is visible analysis rather than hidden preprocessing. Table 8 lists the planned entries; counts are filled by the run.

Table 8. Cleaning ledger (planned entries; counts from the audit run).

ID	File	Problem	Diagnosis	Decision	Rows	Validation
L01	DS Jobs	Salary strings such as 7.8L	Regex parse	Numeric lakh per annum	Not audited in this chart pass	Round-trip format check; $\min \leq \text{avg} \leq \max$
L02	DS Jobs	Repeated reference_no	Field comparison within id	Decision deferred; not audited in this chart pass	Not audited in this chart pass	H1 rerun both ways

ID	File	Problem	Diagnosis	Decision	Rows	Validation
L03	Analytics	Salary formats; undisclosed values	Regex families; unmatched frequency	Parse; undisclosed to missing plus indicator	Not audited in this chart pass	Role mix of disclosed versus undisclosed (chi-squared)
L04	Analytics	Experience ranges	Regex	Min, max, midpoint	Not audited in this chart pass	Reversed and implausible counts
L05	Analytics	job_desig variants	Jaro-Winkler clusters + review	Map to role family	Not audited in this chart pass	Manual audit of 200 mappings
L06	Analytics	key_skills spelling and format	Alias table + fuzzy match	Canonical skills	Not audited in this chart pass	Gold-set F1
L07	JDS, SDS	Spaces in column names	Schema inspection	snake_case in code; originals kept in dictionary	All	Schema test
L08	JDS, SDS	Duplicate or near-duplicate rows	Exact and Hamming match	Decision deferred; not audited in this chart pass	Not audited in this chart pass	Group-aware CV sensitivity

3.4 Derived variables

Table 9. Derived variables.

Variable	Definition	Source
salary_mid_lpa, log_salary	Mean of min and max when both exist, else average; natural log	Both job files
salary_range_ratio	$(\text{max} - \text{min}) / \text{midpoint}$	DS Jobs
exp_mid, exp_width	Midpoint and width of the experience range	Both job files
role_family	Rule-based mapping of title tokens (analyst, scientist, engineer, ML, architect, business, other), reviewed by hand	Both job files
seniority	Ordinal from designation tokens (junior, none, senior, lead, principal, manager)	Both job files
n_skills; dim_share_d	Count of mapped skills; share of a posting's mapped skills in dimension d	Analytics Jobs

Variable	Definition	Source
T, C	T: number of technical dimensions requested (0 to 4). C: 1 if a dashboard-and-storytelling or communication skill is requested	Analytics Jobs
demand_intensity	num_of_jobs (openings) per role family and company	DS Jobs

3.5 Skill normalisation and the gold set

Raw key_skills strings are not canonical skills. The pipeline is: clean, tokenise, alias lookup, fuzzy match, canonical skill, dimension, role association. Each output row keeps the original string, canonical skill, dimension, mapping method and confidence. Four representations are benchmarked in order of sophistication: dictionary and regex; TF-IDF with word and character n-grams; sentence-embedding nearest neighbour against the alias table; and, if its weights can be used offline, an ESCO-based extractor. Because the taxonomy is built for Indian technology postings and ESCO is a European taxonomy, ESCO coverage of tool names is itself a reported finding.

Ground truth does not exist, so we create it: two team members independently label about 250 postings, stratified by role family, and we report Cohen's kappa, then micro and macro precision, recall and F1 with 95% bootstrap intervals, plus coverage (share of tokens mapped). Without the gold set no F1 is claimed.

Table 10. Dimension mapping rules and examples.

Dimension	Example canonical skills	Ambiguity rule
Big data	Hadoop, Spark, Hive, Kafka, data warehousing	Cloud data platforms included; SQL excluded here
Mathematics and statistics	Statistics, regression, probability, hypothesis testing, forecasting, optimisation	Generic 'analytics' not mapped
Coding	Python, Java, Scala, C++, R, SQL	SQL assigned here; sensitivity run reassigns to big data
AI and ML	Machine learning, deep learning, NLP, computer vision, TensorFlow, PyTorch	Generic 'AI' with no method named is mapped
Dashboard and storytelling	Tableau, Power BI, data visualisation, reporting, presentation, stakeholder communication	Communication is counted in C as well
Unmapped	ERP, CRM, project tools, domain systems	Counted and reported; excluded from T and C

3.6 Leakage audit

Table 11. Leakage tests. No benchmark is accepted unless every row is PASS or an explained WARN.

Test	What it catches	Mechanism	Verdict
LK1	ID or index columns used as features	Feature allow-list assertion	Not audited in this chart pass
LK2	Duplicate rows straddling outer folds	Group-aware split check	Not audited in this chart pass
LK3	Preprocessing fitted outside the training fold	Pipeline objects only; code assertion	Not audited in this chart pass
LK4	Target-derived or deterministic features	Feature-to-target mapping and correlation screen	Not audited in this chart pass
LK5	Calibration data overlapping the scored fold	Inner out-of-fold calibration only	Not audited in this chart pass
LK6	Outcome information reaching the taxonomy	M2 never reads JDS or SDS	Not audited in this chart pass
LK7	Harness bias	Shuffled-target AUC centred on 0.5	Not audited in this chart pass
LK8	Stacking leakage	Meta-learner fitted on out-of-fold predictions only	Not audited in this chart pass

3.7 Exploration figures

Exploration is organised by hypothesis, not by chart type. Figures F5 to F9 (Appendix B) answer, in order: how salary varies by role family and experience, which skills dominate and for whom, how missingness and undisclosed salary are structured, and what the JDS and SDS marginals and correlations look like. The missingness figure is shown first because undisclosed salary can bias every salary result.

Figures F5 and F6 are included in Appendix B.

Figure F8 is included in Appendix B.

4. Data Analysis

Each subsection follows one pattern: hypothesis, method, expected form of result, limitation. Figure slots point to the specifications in Appendix B.

4.1 Market structure (H1)

For DataScience Jobs we test whether log salary differs across role families with Kruskal-Wallis (rank-based, robust to the right skew of salary), report epsilon-squared with a bootstrap interval, and run Dunn's pairwise tests with Holm correction. Because company names repeat, we also fit $\log_salary \sim \text{role_family} + \text{exp_mid} + (1 | \text{company})$ and report the share of variance at company level. Demand is analysed through num_of_jobs per role family, and the two are combined in the Career Opportunity Frontier (F11): median salary on one axis, openings on the other, bubble area for median minimum experience, colour for role family, with bootstrap ellipses so that apparent differences are shown with their uncertainty. Limitation: the postings are a sample of advertised roles, not of employment.

4.2 Skill structure (H2)

For each skill s and role family r we compute the log-odds ratio of s in r against all other roles with an informative Dirichlet prior (Monroe, Colaresi and Quinn), which shrinks rare-skill estimates toward the corpus rate, and convert it to a z-score. The null is built by permuting role labels 1,000 times; H2 is supported if the number of skills with FDR $q < 0.05$ clearly exceeds the null. The co-occurrence graph weights edges by normalised pointwise mutual information (NPMI), which corrects for the dominance of very frequent skills, and is clustered with Leiden at several resolutions. Because no ground truth exists for a skill graph, we validate it as the software-skills literature does: mask 40% of a posting's skill entries and test whether graph relatedness recovers them better than a frequency-only baseline (F9). Cluster stability is the adjusted Rand index across 200 bootstrap resamples; a clustering with unstable assignments is reported as unstable.

The **Skill Market Signal Index** is an analytic ranking, not a value of a skill. For skill s , $SSI(s) = \sum_k w_k \cdot r_k(s)$, where r_k is the percentile rank on four components (demand, role specificity, salary association, breadth across role families) and the salary association is the ridge-shrunk coefficient of a skill indicator in log salary controlling for role, experience and location. Weights are not chosen: we draw w from $\text{Dirichlet}(1,1,1,1)$ 10,000 times and report each skill's rank interval and the Kendall tau against equal weights (F10).

Figures F7 and F9 are included in Appendix B.

4.3 Complementarity in postings (H3)

We fit log salary on role family, experience midpoint, location, T, C and the product $T \times C$, with heteroscedasticity-robust standard errors, and repeat at the 0.25, 0.5 and 0.75 quantiles to see whether a premium exists across the distribution rather than only at the mean. The disclosed-salary subsample may differ from all postings, so the model is also fitted with inverse-probability weights from a disclosure model, and the two estimates are shown together. Robustness variants: binary T, dimension shares in place of counts, the random SQL reassignment from Section 2.3, and exclusion of flagged outliers.

4.4 Junior lane (H4)

The target is the high/low hike label with five scores as features. Candidates are a majority-class baseline; logistic regression; Firth-penalised logistic regression; logistic regression with the storytelling $\times$ maths-statistics interaction; random forest; extra trees; histogram gradient boosting; LightGBM; EBM; and TabPFN where its weights can run offline. Hyper-parameters are tuned in the inner loop (Appendix D). H4 is tested with a likelihood-ratio test of the interaction model against the additive model, a 2,000-replicate bootstrap interval for the coefficient, and a simulation that reports the smallest interaction effect detectable with 80%

power at $n = 139$, so that a null result is reported as low power and not as absence of effect. Metrics are ROC-AUC, PR-AUC, F1, balanced accuracy, accuracy, Brier score and bootstrap ECE, averaged over the 100 outer folds with standard deviations.

4.5 Senior lane (H5)

The SDS analysis begins with forensics (Table 7, last row) because a preliminary AUC near 0.992 on five features and 161 rows is unusual for behavioural data. We fit a depth-2 and depth-3 decision tree and look for a simple threshold rule; compute the shuffled-target gap; and inspect the trait marginals for ceiling effects. If a simple rule separates the classes, the result is reported as a property of how the label relates to the trait scores in this file, not as evidence about real-world success. We then compare additive logistic regression, logistic regression with quadratic and pairwise terms (limited to conscientiousness $\times$ openness, conscientiousness $\times$ extraversion and openness $\times$ extraversion, chosen in advance), EBM and a depth-3 tree. The non-additive hypothesis has a specific, literature-based prediction: studies of the Big Five and job performance report curvilinear effects of conscientiousness and interactions among factors, so we examine partial-dependence and EBM shape curves for a plateau or decline at high conscientiousness. Limitation: those studies use supervisor-rated performance, whereas the SDS label is a high/low success classification of unstated provenance.

4.6 Ensembles, calibration and uncertainty

Stacking is attempted with base learners from the logistic, tree and boosting families, using only out-of-fold predictions for the L2-regularised logistic meta-learner. It is kept only if the paired Δ AUC against the best single model has an interval that excludes 0, using the corrected resampled t-test of Nadeau and Bengio, which accounts for the overlap between training sets in repeated cross-validation. Probabilities are compared uncalibrated and Platt-calibrated; isotonic regression is not used because it overfits at these sample sizes. Reliability diagrams use five bins with a bootstrap band, and ECE and Brier score are reported with intervals. For per-case uncertainty we use split-conformal prediction sets (Angelopoulos and Bates) at $\alpha = 0.10$ and report empirical coverage and the distribution of set sizes; with n this small the coverage itself has sampling error, which is shown.

Figures F13 to F16 are included in Appendix B.

4.7 Explanation

We report permutation importance computed on the sealed fold, SHAP values for tree models, EBM shape functions with bagged intervals, and the coefficients of the linear and Firth models, and require that the four agree in sign and rough rank before a feature is called important. Explanations are written as contributions to the model's estimate of the observed class, never as effects of changing a person's score. Auditor rule A3 enforces that wording.

Figures F17 and F18 are included in Appendix B.

4.8 Ablation and robustness

Table 12. Ablation plan; each row is a pre-registered comparison reported with an interval.

Lane	Variants
JDS	A all five features; B technical only (big data, coding, AI and ML); C communication only (dashboard and storytelling); D quantitative only (maths-statistics, coding, AI and ML); E drop the strongest feature; F shuffled target
SDS	A all five traits; B without neuroticism; C conscientiousness only; D openness only; E with the three pre-chosen interactions; F shuffled target
Skills	A raw keyword match; B TF-IDF; C embeddings; D taxonomy-assisted matching; each scored on the gold set
System	Stacking versus best single model; calibrated versus raw; TabPFN and EBM versus tuned logistic

Robustness covers 20 random seeds, three cross-validation geometries (5×20, 10×10, 3×30), decision-threshold changes, missing-value perturbation, Gaussian noise on features at 0.25 and 0.5 SD, exclusion of flagged outliers and near-duplicates, and the 200-permutation shuffled-target null. If performance collapses under any perturbation, the collapse is documented in the results register.

4.9 Demand-reward alignment (descriptive)

For each of the five dimensions d we compute, in postings, the share of postings requesting d by role family (with a bootstrap interval over postings), and, in JDS, the standardised log-odds of a high hike per one-SD increase in the dimension score (bootstrap over respondents, other dimensions held fixed in the multivariable model). We plot the two as a pair per dimension with a 95% joint interval (F12) and report how often, across 2,000 joint bootstrap draws, a dimension's demand rank and reward rank differ by two or more places. The posting and JDS populations are different, and so are their outcomes (advertised salary versus realised hike), so the comparison is an alignment gap and not a measure of mispricing.

Figures F11 and F12 are included in Appendix B.

5. Results and Conclusions

5.1 Preliminary benchmark

Figure 4 retains the earlier single-seed benchmark for historical context. Table 13 and Figures F13–F20 report an exploratory rerun from the supplied data using 20 repeats of 5 stratified folds and fixed model settings. These results do not use the proposed nested hyper-parameter search and should not be read as final model-selection claims.

Table 13. Exploratory repeated-CV benchmark using fixed model settings.

Dataset	Model	n	Mean ROC-AUC	2.5–97.5% fold quantiles
JDS	Logistic	139	0.909	0.792–0.992
JDS	Extra Trees	139	0.848	0.710–0.954
JDS	HistGB	139	0.873	0.749–0.980
SDS	Logistic	161	0.964	0.900–1.000
SDS	Extra Trees	161	0.997	0.982–1.000

SDS	HistGB	161	0.992	0.965–1.000
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Preliminary model benchmark (not final; see validation protocol, Figure 3)



Figure 4. Preliminary ROC-AUC and accuracy by model (single-seed, five-fold exploration; retained for historical comparison).

Two patterns in these numbers define the questions of Section 4. In JDS the linear model leads and the flexible models trail it (extra trees 0.753 versus logistic 0.903), which is what an approximately additive signal plus noise at $n = 139$ looks like. In SDS the ordering reverses (logistic 0.949 versus extra trees 0.998), a gap that implies either non-additive structure or rule-like label generation. H5 and the forensic tests exist to tell these apart. Univariate associations in the same exploratory pass (Table 14) are hypotheses, not findings.

Table 14. Preliminary univariate associations with the outcome (approximate; hypothesis-generating).

JDS feature	Association	SDS trait	Association
Dashboard and storytelling	≈ 0.55	Conscientiousness	≈ 0.68
Maths and statistics	≈ 0.52	Openness	≈ 0.67
Coding	≈ 0.44	Extraversion	≈ 0.49
AI and ML	≈ 0.40	Agreeableness	≈ 0.29
Big data	≈ 0.11	Neuroticism	≈ 0

5.2 Results register

The exploratory figure pass does not populate the hypothesis register. H1–H5 remain untested here or pending audit; Table 13 reports only the exploratory classification benchmark.

Table 15. Results register.

ID	Primary estimate	95% interval	Verdict	n	Claim ids
H1	Not estimated in this exploratory chart pass	Not estimated	Untested in this chart pass	1,602 postings	No validated claim ID
H2	Not estimated in this exploratory chart pass	Not estimated	Untested in this chart pass	15,841 postings	No validated claim ID

ID	Primary estimate	95% interval	Verdict	n	Claim ids
H3	Not estimated in this exploratory chart pass	Not estimated	Untested in this chart pass	15,841 postings	No validated claim ID
H4	0.589 (source registry; pending audit)	Not verified	Pending audit	139 records	H4-int-001
H5	Not estimated in this exploratory chart pass	Not estimated	Untested in this chart pass	161 records	No validated claim ID
Align	See exploratory Figure F12	Descriptive intervals	Descriptive only	15,841 postings + 139 JDS records	No validated claim ID

5.3 Pre-committed interpretation

To prevent post-hoc storytelling, the reading of each outcome pattern is fixed here.

Table 16. Outcome patterns and the conclusion each will support.

Pattern	Conclusion we will state	Wording limit
H3 and H4 supported with the same sign; H5 shows a conscientiousness-by-social interaction	Technical and social skill complementarity appears in three independent files	Association only
H3 supported, H4 null	A salary premium for the pairing is visible in postings; JDS at n = 139 cannot confirm it (state MDE)	Do not call the JDS result absence of effect
H3 null, H4 supported	Progression signal in JDS is not visible in advertised salary: an alignment gap	Populations and outcomes differ
All null	No evidence of complementarity in these data; an additive skill model suffices	Report power
A simple rule separates SDS classes	The SDS label is nearly a function of the trait scores in this file; findings describe the labelling	No workforce inference

5.4 What the numbers cannot say

Cross-validated performance estimates how well a model fits the same population it was sampled from. It does not show deployment performance, and ROC-AUC is not accuracy: both are always reported by name. A feature with large importance contributes strongly to the model's estimate of the observed class; it is not shown to cause the outcome.

6. Implications

Implications are conditional on the outcome pattern in Table 16 and are limited to what the evidence supports.

Table 17. Stakeholder implications.

Stakeholder	If the association is supported	If null	Not supported by this evidence
Employers	Benchmark pay within role family, not pooled; check whether postings name communication alongside technical requirements	Pooled technical skill lists are adequate	Screening individuals on personality scores
HR and L&D	Programmes pairing quantitative training with visual and verbal communication are the better-supported bundle	No evidence for a bundle effect; prioritise by role-specific demand	Predicting any employee's promotion
Universities	Compare dimension demand shares with curriculum coverage (needs institutional data)	Align to role-specific skills from H2	Claims about graduate outcomes
Professionals	Use role-conditioned skill gaps and adjacent-skill clusters to plan learning	Prioritise the most role-specific skills	Guaranteed salary gains from any skill
Policymakers	Invest in longitudinal linkage with consent; localise skill taxonomies for the Indian market	Same data need	Causal policy conclusions

7. Limitations and Ethical Considerations

L1 Observational data. No causal inference is made. **L2 Small samples.** JDS (139) and SDS (161) give wide intervals and low power for interactions, so every interaction result carries a minimum detectable effect. **L3 Provenance.** The files may contain self-reported, masked or generated values, and the SDS label's provenance is unstated. **L4 No linkage.** There is no person-level link between files, so alignment is across populations. **L5 Construct validity.** Mapping posting skills onto the five JDS dimensions is our construct; it is tested for sensitivity but not validated against JDS item content. **L6 Taxonomy fit.** External taxonomies may fit Indian technology postings poorly. **L7 Sampling.** Postings are not the whole market and undisclosed salaries may be selected. **L8 No time axis.** No trend or trajectory is observed. **L9 Personality.** Trait scores are normalised self-descriptions; interpretations must be cautious. **L10 Generalisation.** Cross-validation does not show real-world performance.

Ethics. We do not build or recommend a tool that scores individuals. The output of any model is an estimate of the observed class under a learned statistical relationship, and personality results in particular must not be used to screen, rank or exclude people. The auditor's language rule blocks deterministic and discriminatory phrasing in generated text.

8. Conclusion and Future Work

We defined a problem the four files can support, built the machinery to test it, and fixed in advance how each outcome will be read. The contribution is the composition: a falsifiable organising hypothesis tested in three independent datasets, a boundary-preserving architecture in which datasets meet only at shared constructs, and an audit layer that makes every number traceable. None of the individual techniques is new, and we do not present them as such. Future work, in order of value: linked longitudinal data collected with consent; dated postings for genuine trend analysis; documented JDS items to validate the bridge; a localised skill taxonomy for Indian postings; and causal designs (for instance, matched cohorts) that observational files cannot support.

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- [12] Using digital traces to analyze software work: skills, careers and programming languages. arXiv:2504.03581.

Appendix A. Data Dictionary (as supplied)

File	Column	Meaning and handling
DataScience Jobs	reference_no	Posting reference; not a feature; repeated values investigated (L02)
	company_name, job_title	Categorical; normalised (Table 7)
	min_experience	Years; parsed
	avg_salary, min_salary, max_salary	Strings such as 7.8L; lakh per annum after parsing
	num_of_jobs	Openings; demand intensity
Analytics Jobs	s_no	Row id; not a feature
	experience, salary	Free-text ranges; parsed
	job_description, job_desig, job_type, location	Text and categorical; NLP and category normalisation
	key_skills	Delimited skill strings; normalised to canonical skills
JDS	id	Row id; not a feature
	big_data_skills, maths-stats_skills, coding_skills, ai_and_ml_skills, dashboard_and_storytelling_skills	Scores 1 to 5
	salary_hike_high_or_low	Outcome: 1 high, 0 low
SDS	id	Row id; not a feature
	neuroticism, extraversion, openness_to_experience, agreeableness, conscientiousness	Normalised trait scores; column names with spaces normalised in code, originals preserved here
	success_classification_high_low	Outcome: high or low success

Appendix B. Figure Specifications

Table B1. Figure specifications. Figures F5–F20 are generated in this revision from the supplied files; interpretation limits are stated below.

ID	Figure	Encoding	Annotation or test
F1 to F4	Architecture, no-false-join, validation protocol, preliminary benchmark	As in Sections 2 and 5	Done
F5	Salary by role family	Violin plus median and IQR, log LPA	n per family; Dunn-Holm brackets
F6	Experience versus salary	Hexbin plus LOESS per role family	Bootstrap bands

ID	Figure	Encoding	Annotation or test
F7	Role-skill heatmap	30 skills by 8 role families, colour = log-odds z	Hatching where $q \geq 0.05$
F8	Missingness and disclosure	Missingness matrix; role mix bars	Chi-squared
F9	Masked-skill validation	ROC and lift by decile	Versus frequency baseline
F10	Signal Index rank intervals	Dot plot of rank interval for top 25 skills	10,000 Dirichlet weight draws
F11	Career Opportunity Frontier	x openings, y median log salary, bubble experience, colour role	Bootstrap ellipses
F12	Demand-reward alignment	Per dimension: demand share against JDS log-odds	95% joint ellipse
F13	Benchmark leaderboard	Forest plot of AUC over 100 outer folds, JDS and SDS	Shuffled-target null band
F14	ROC and PR curves	Per-fold light lines with mean	Per model
F15	Reliability diagrams	Five bins before and after Platt	Bootstrap band; ECE, Brier
F16	Conformal coverage	Coverage versus $1 - \alpha$; set-size histogram	$\alpha = 0.10$
F17	Shape functions	EBM shapes with bagged intervals; PDP	Conscientiousness plateau check
F18	Interaction surfaces	Probability heatmaps for two pairs	Convex hull of data marked
F19	Ablations	Bars with intervals per Table 12	Paired Δ AUC
F20	Robustness	AUC across seeds, splits and noise levels	Collapse points flagged

Generated figures from the supplied data

Data sources: DataScience Jobs.csv (n=1,602), Analytics Jobs.csv (n=15,841), JDS Skill Traits.xlsx (n=139), and SDS Personality Traits.xlsx (n=161). The descriptive posting figures use reproducible keyword rules for role families and skill dimensions; these rules are exploratory and do not replace the planned curated taxonomy or hand-labelled gold set. F9 uses synthetic masked tokens, and F10 uses row-bootstrap rank intervals rather than the planned taxonomy-weighted Dirichlet index. F13–F20 use repeated 20×5 stratified folds with fixed settings; no nested tuning, EBM, TabPFN, hand-labelled gold set, or original analysis code was available. Calibration intervals are binomial approximations; model outputs remain exploratory. Fold quantiles describe split variability, not independent-sample confidence intervals. Cross-file alignment is descriptive, with no row-level joins or causal interpretation.

Figure F5. Salary distributions by exploratory title family in DataScience Jobs.

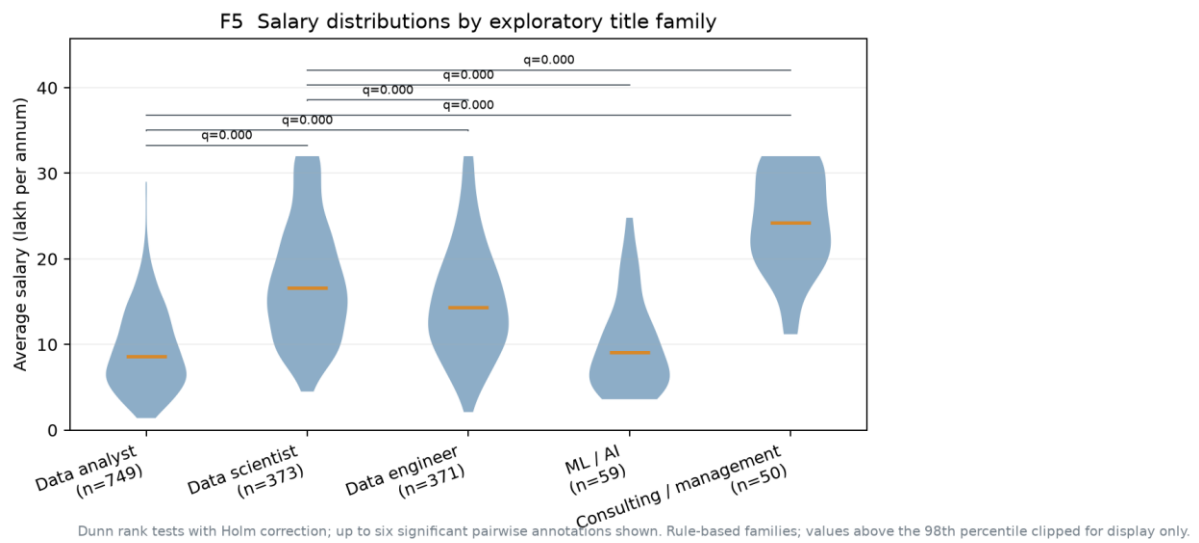


Figure F6. Experience and advertised salary with LOWESS trends and bootstrap bands.

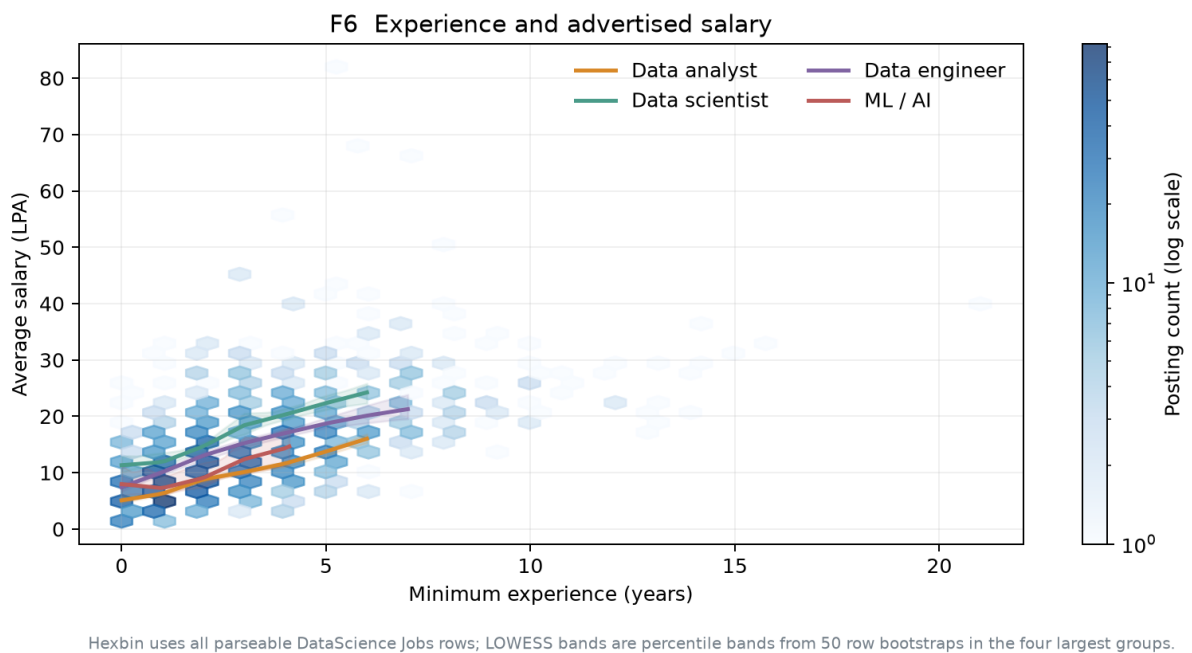


Figure F7. Role-skill association heatmap for frequent raw skill strings in Analytics Jobs.

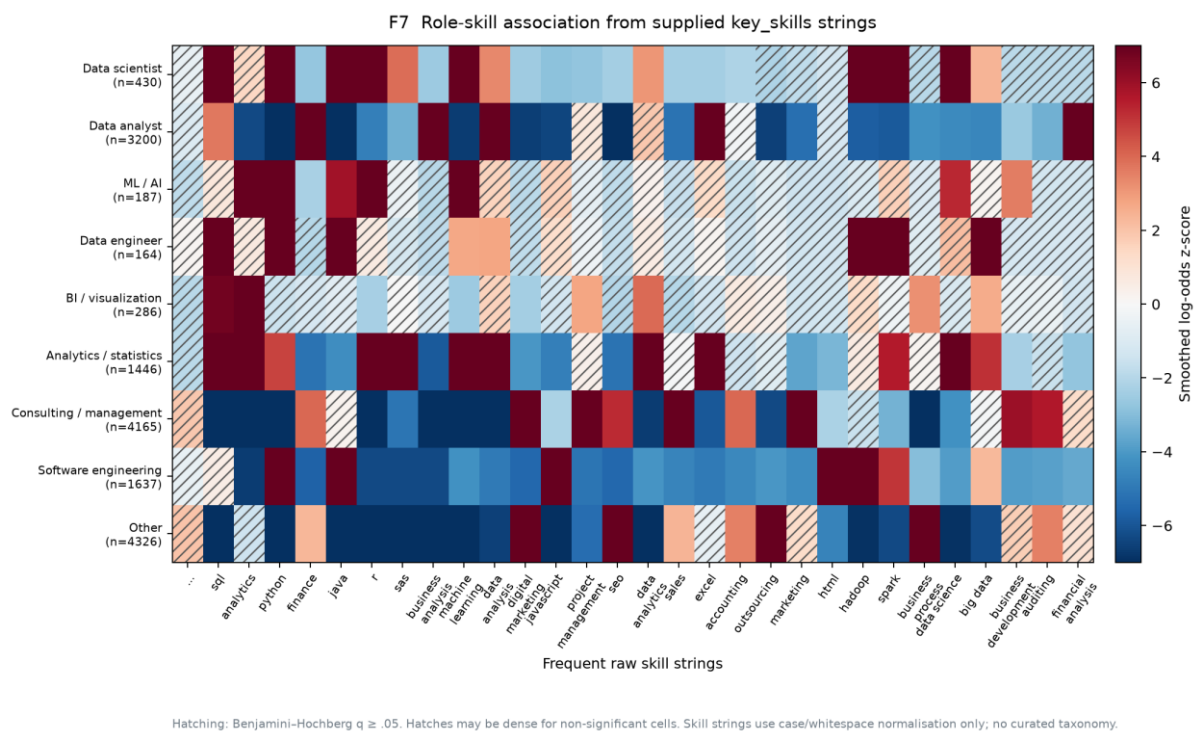


Figure F8. Missingness matrix and salary disclosure by exploratory role family.

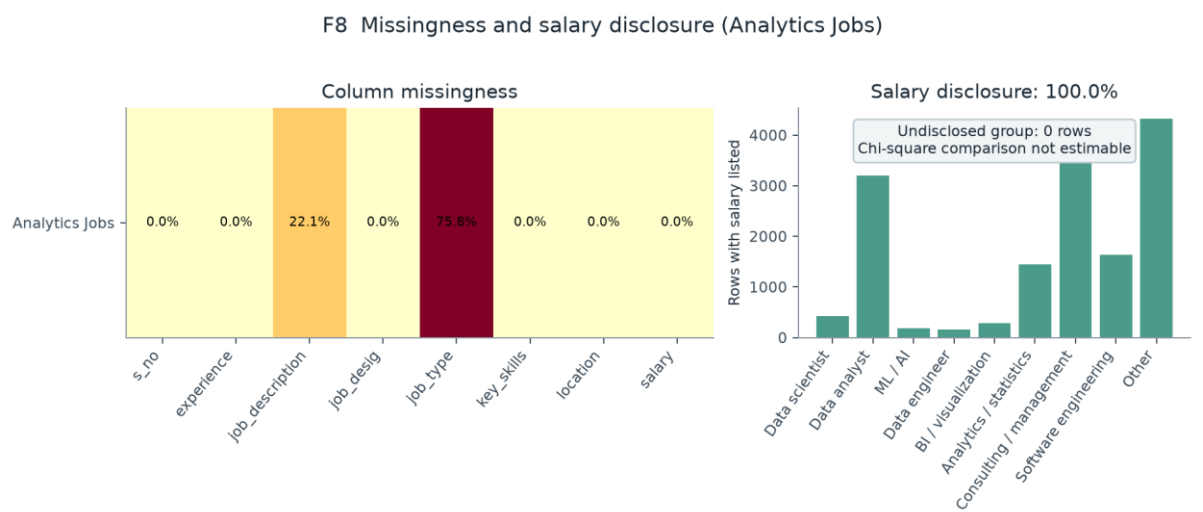


Figure F9. Synthetic masked-skill retrieval proxy compared with a frequency baseline.

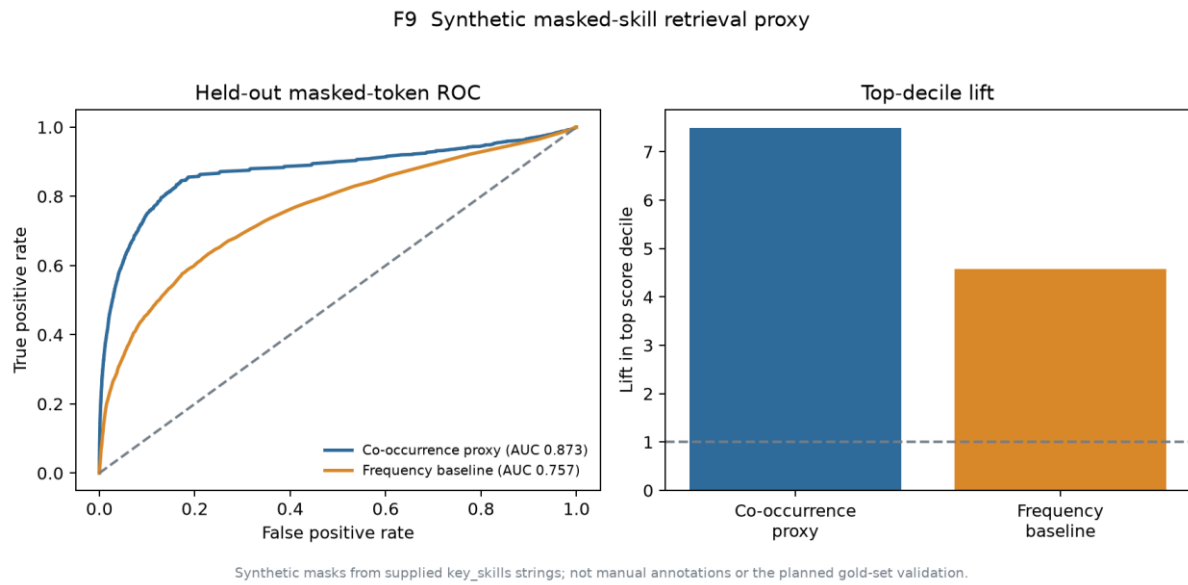


Figure F10. Bootstrap rank intervals for the 25 most frequent raw skill strings.

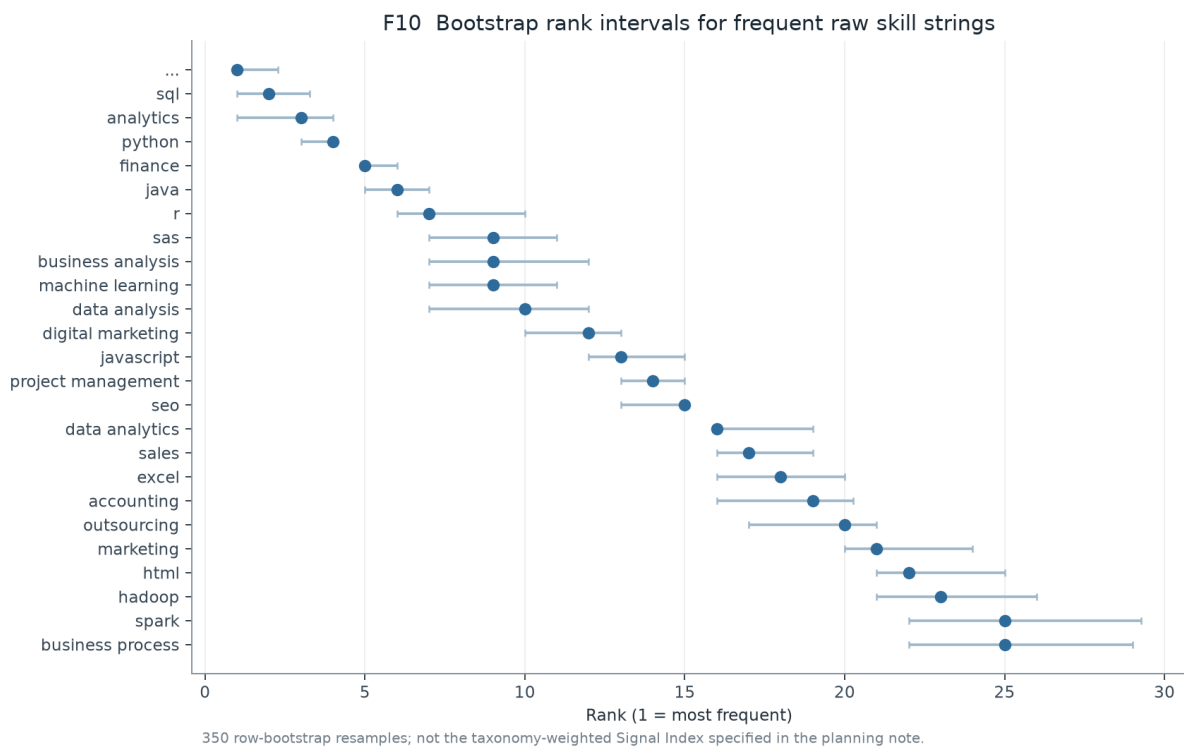


Figure F11. Career opportunity frontier by exploratory title family.

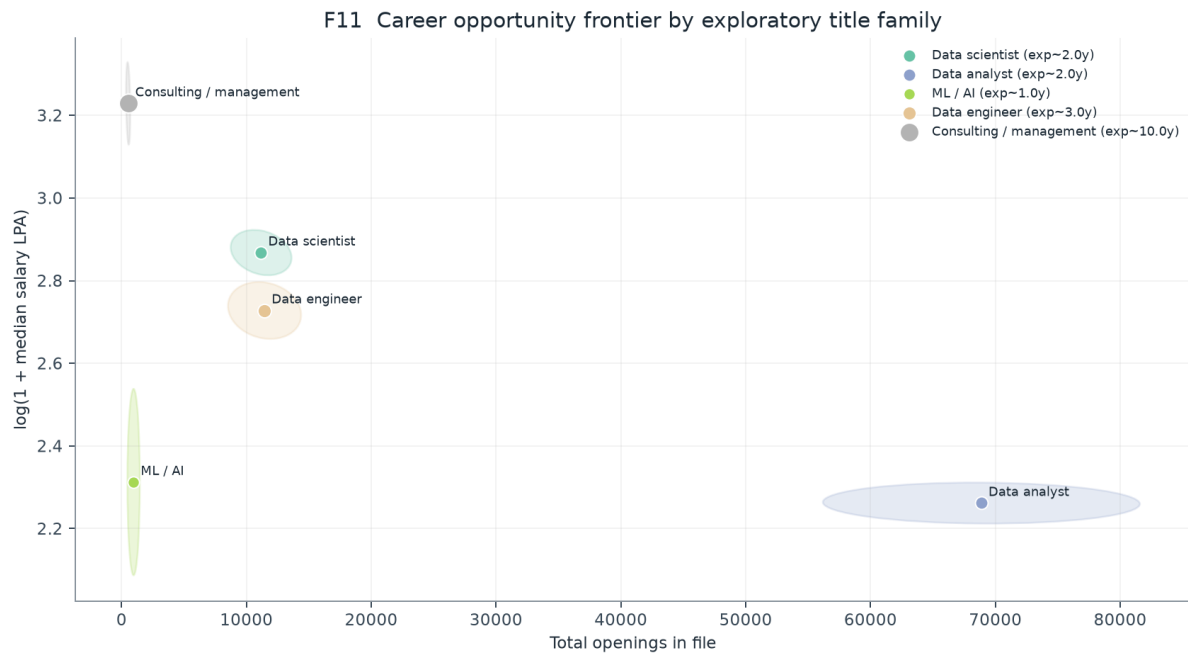
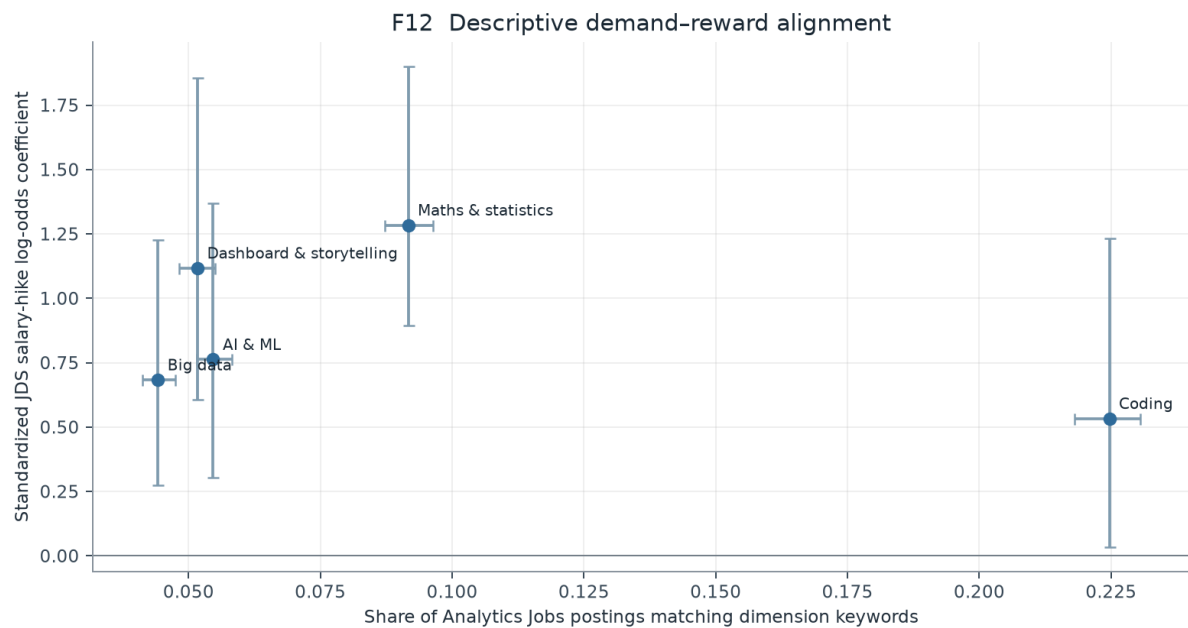


Figure F12. Descriptive alignment between keyword-mapped demand and JDS skill associations.



Mapping is exploratory keyword matching; cross-file comparison is descriptive, not a linked-person or causal estimate.

Figure F13. Repeated-CV leaderboard with a shuffled-target reference band.

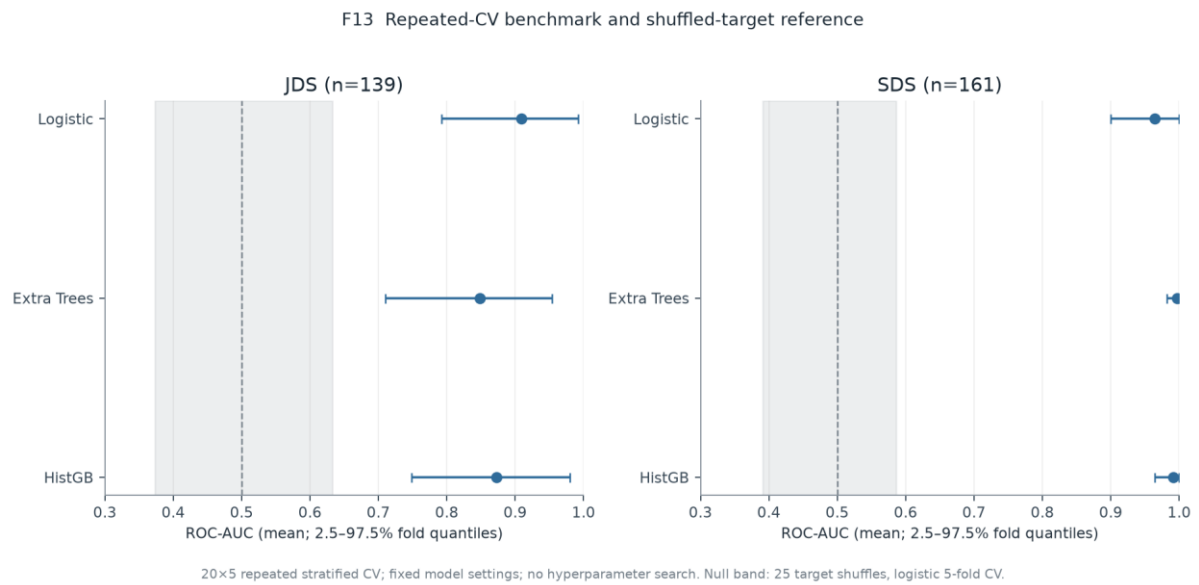


Figure F14. Mean ROC and precision-recall curves across repeated test folds.

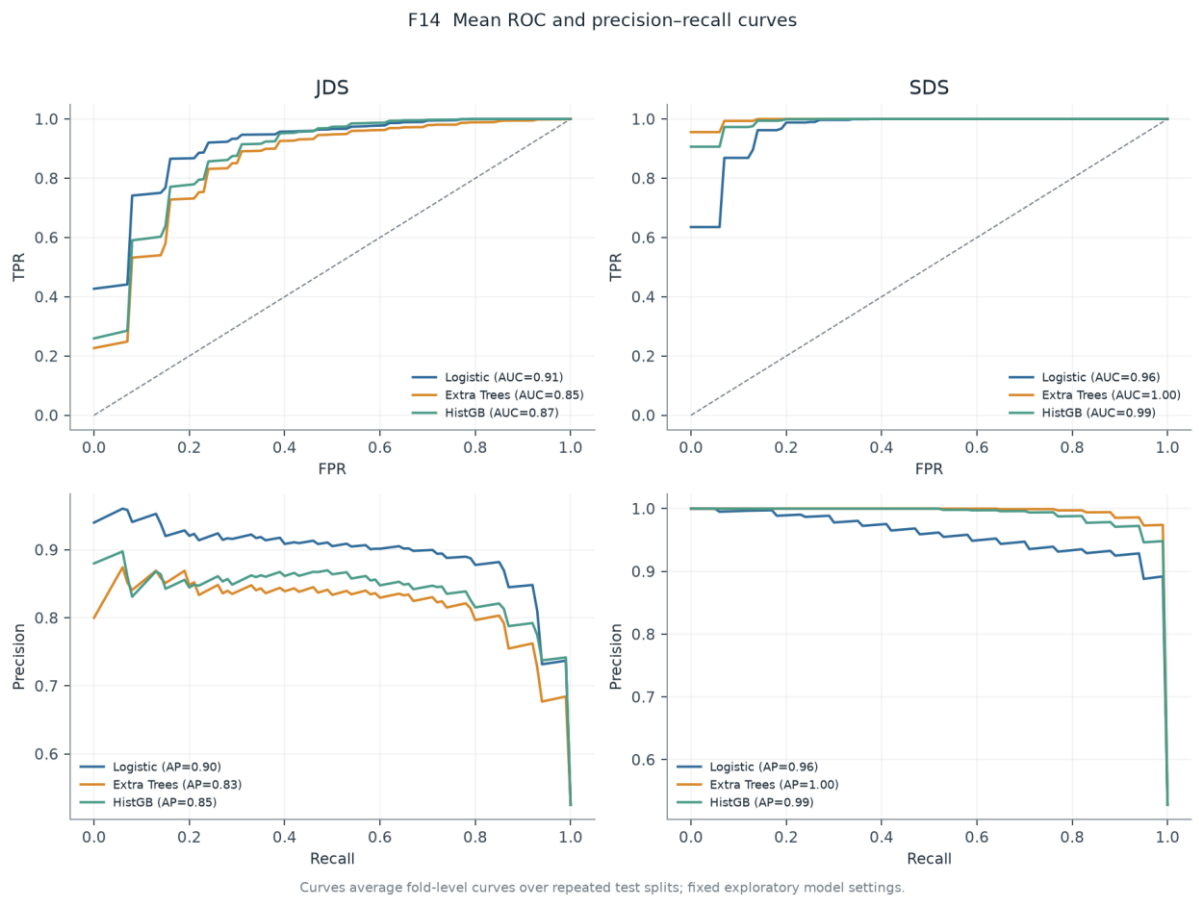


Figure F15. Reliability before and after Platt calibration.

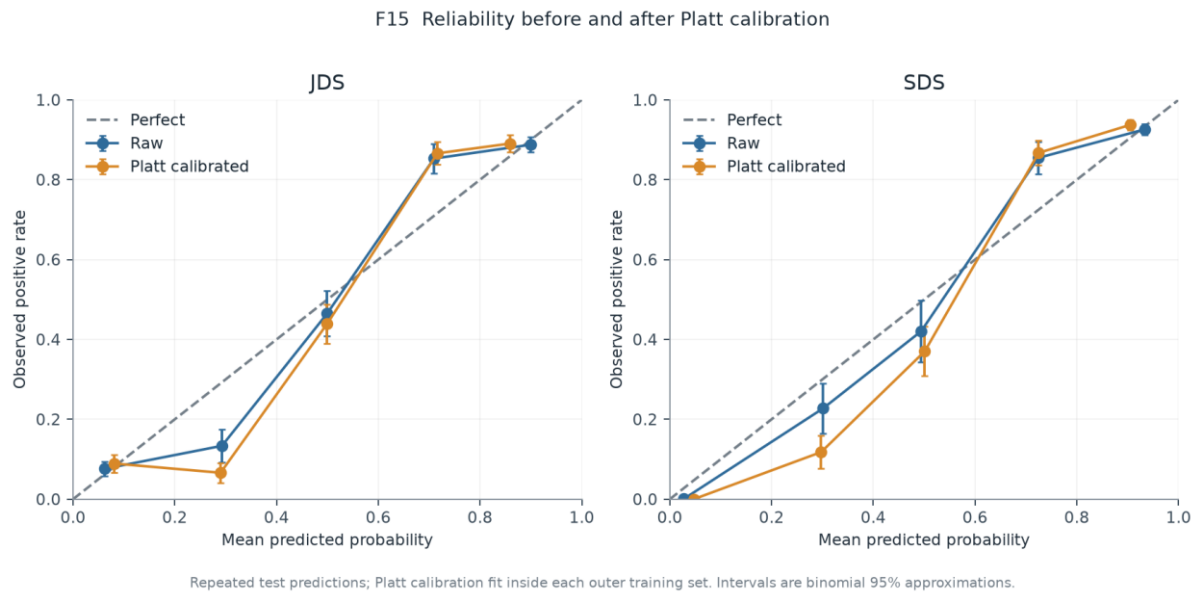


Figure F16. Split-conformal coverage and mean prediction-set size.

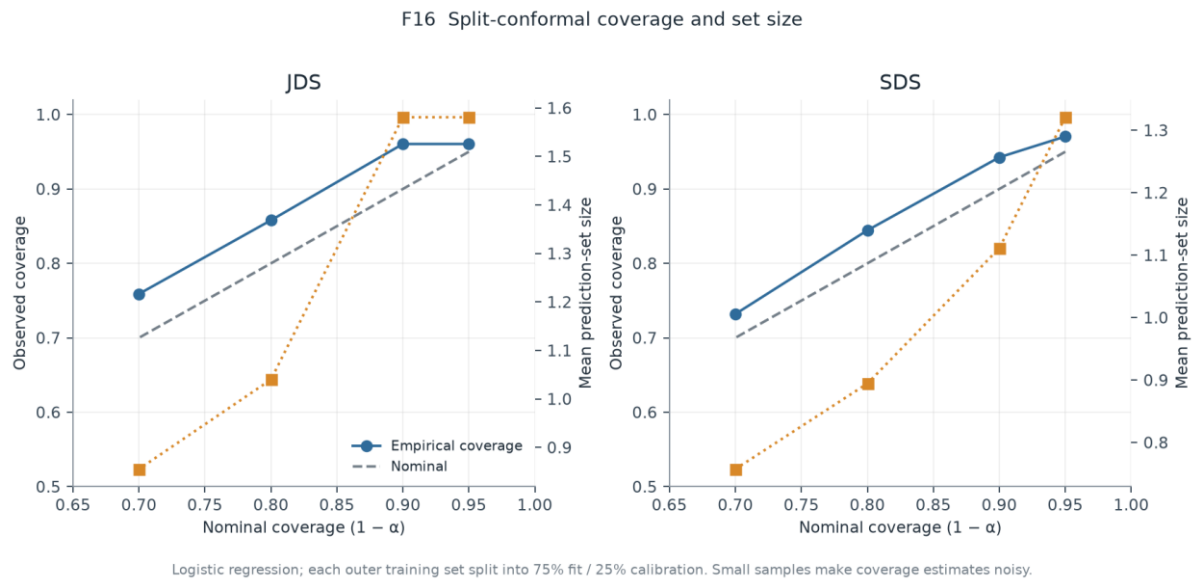
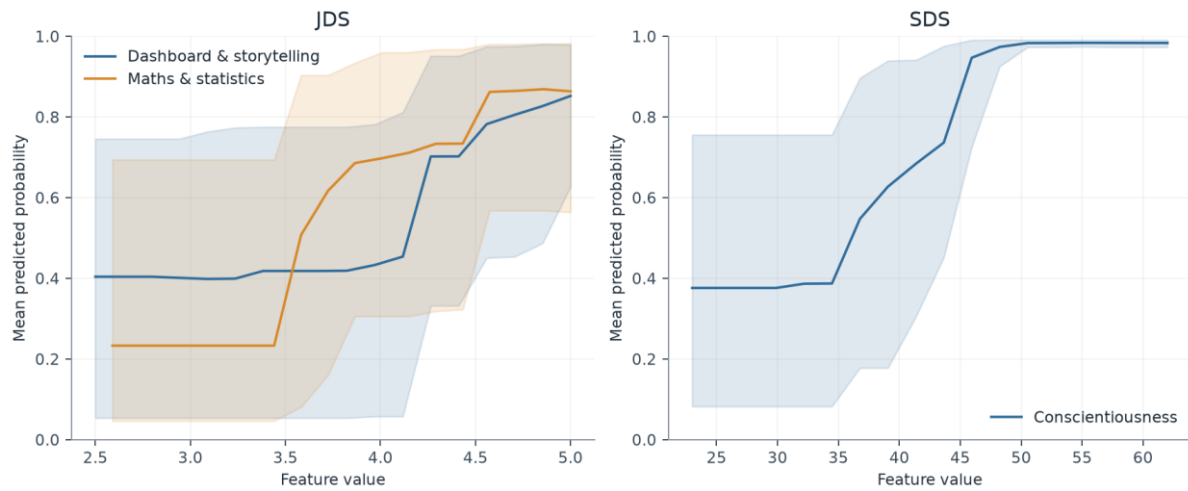


Figure F17. Partial-dependence shapes with respondent-bootstrap bands.

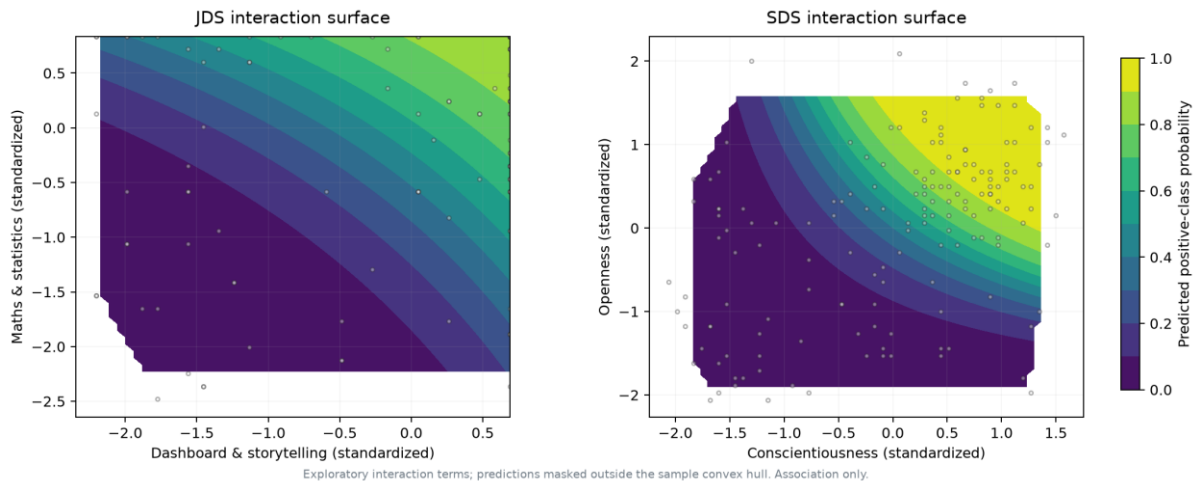
F17 Partial-dependence shapes with bootstrap bands



HistGradientBoosting partial-dependence proxy; 50 respondent bootstraps. EBM software and the original fit are not supplied.

Figure F18. Logistic interaction surfaces inside the observed convex hull.

F18 Logistic interaction surfaces inside observed convex hull



Exploratory interaction terms; predictions masked outside the sample convex hull. Association only.

Figure F19. Feature ablations with paired fold-level AUC changes.

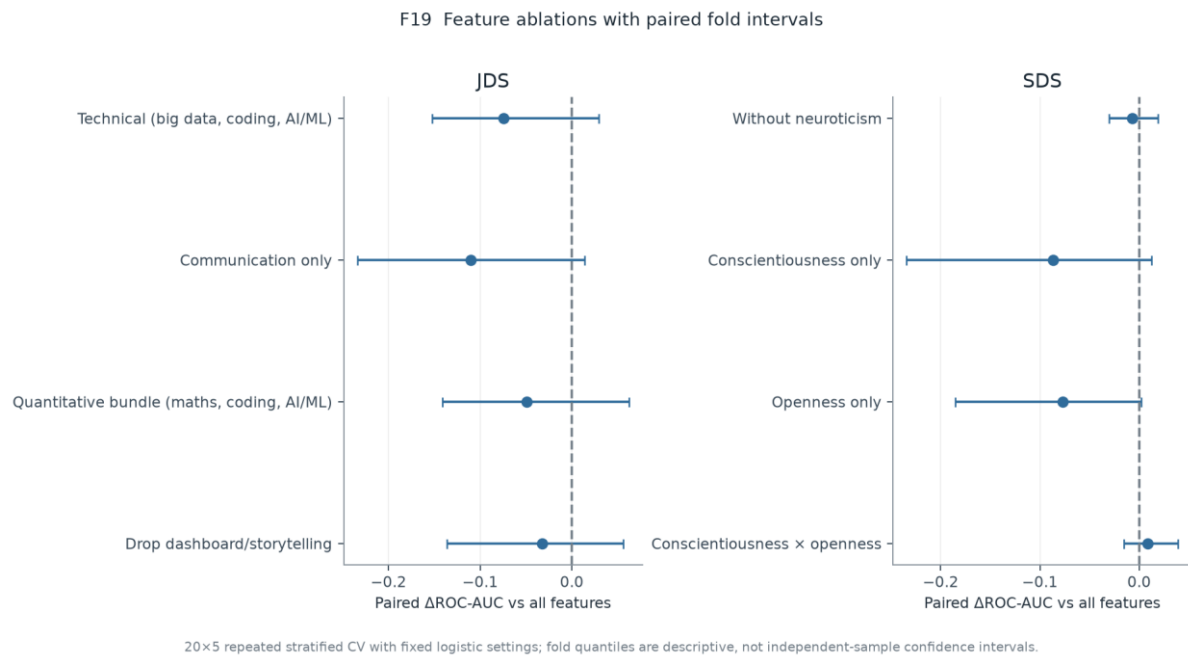
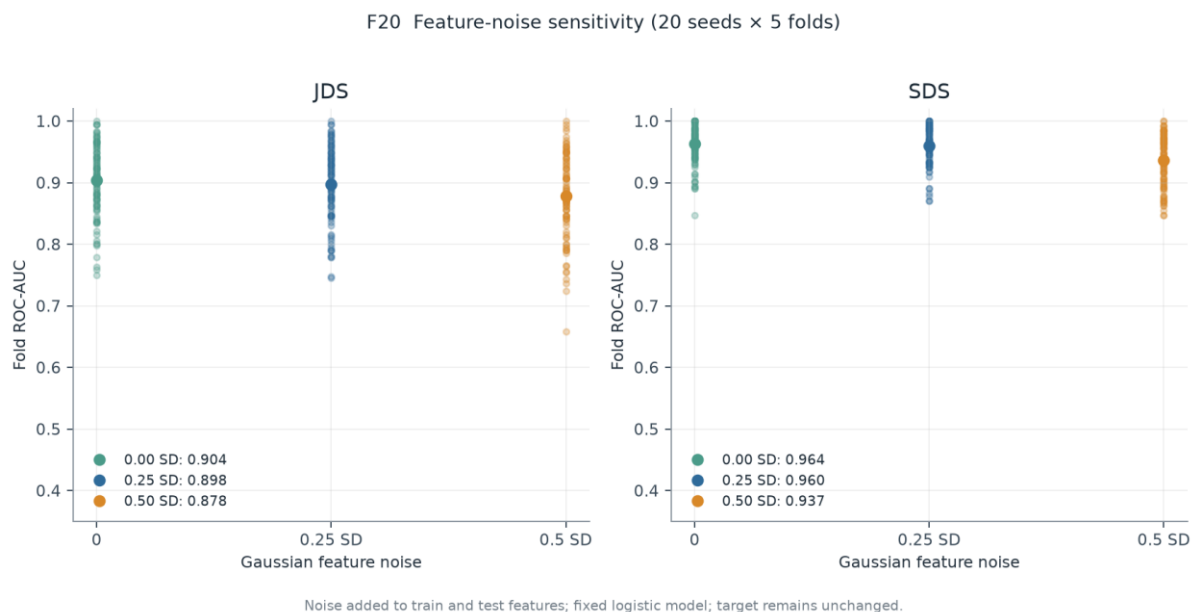


Figure F20. Feature-noise sensitivity across repeated stratified folds.



Appendix C. Evidence Registry Record

claim_id: H4-int-001

statement: "Adding the storytelling x maths-statistics interaction changes JDS hike log-odds"

dataset: JDS (tag O/M)

method: Firth logistic; LRT additive vs interaction; 2,000-rep bootstrap

metric: interaction coefficient (log-odds per SD^2); delta-AUC (nested CV)

value: 0.589

ci95: 0.589

n: 139

limitation: "n = 139; MDE = 0.589; association only"

code_ref: src/models/jds_interaction.py @ a1b2c3d4

status: pending_audit

Appendix D. Hyper-parameter Search Spaces (inner loop)

Model	Search space
Logistic (L2)	C in 13 log-spaced values from 1e-3 to 1e3
Random forest, extra trees	500 trees; max_depth {2, 3, 4, none}; min_samples_leaf {1, 2, 4, 8}; max_features {sqrt, 1.0}
HistGB	learning_rate {0.03, 0.1}; max_depth {2, 3}; max_iter {100, 300}; l2 {0, 1, 10}
LightGBM	num_leaves {4, 8}; min_child_samples {5, 10, 20}; learning_rate {0.03, 0.1}; n_estimators {100, 300}; reg_lambda {0, 1, 10}
EBM	interactions {0, 3, 5}; outer_bags 25; learning_rate 0.01
TabPFN	Defaults; no tuning
Stack meta-learner	L2 logistic, C {0.1, 1, 10}

Appendix E. Reproducibility

- One command (make all) reruns audit, features, benchmark, registry and note rendering from the raw files.
- Exploratory chart pass used Python 3.12, pandas 3.0.6, scikit-learn 1.9.1, SciPy 1.18.1 and matplotlib 3.11.2. The original analysis repository, pinned requirements and run log were not included with the supplied files.
- No hidden notebook state: notebooks only display registry outputs.
- Code excerpts and screenshots of the pipeline run: Reproducibility logs and screenshots are available in the repository..

